MEETING INTELLIGENCE PLATFORM

Technical Architecture & Product Design Document

Prepared by	Utilitarian Labs
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01 Executive Summary

The Meeting Intelligence Platform (MIP) is an enterprise-grade, AI-powered system that automatically captures, transcribes, diarizes, summarizes, and distributes meeting content from Google Meet and Microsoft Teams. It operates silently in the background on employees' Windows machines via a lightweight desktop agent, requiring zero manual intervention once deployed.

The platform consists of two primary surfaces: an Admin Panel for IT and HR teams to onboard employees and manage policy, and a Desktop Agent (.exe) deployed on each employee machine that monitors meeting sessions, captures audio, and orchestrates the full AI pipeline.

> ✓ **BEST PRACTICE** Primary goal: Eliminate manual note-taking and follow-up overhead. Every meeting — regardless of platform — is automatically captured, transcribed in English and Hindi, speaker-identified, summarized with action items, and emailed to all attendees within minutes of the meeting ending.

Key Capabilities at a Glance

Capability	Description
Auto-Detection	Detects Google Meet and Microsoft Teams sessions without manual triggers
Bilingual Transcription	Supports English + Hindi using Whisper Large V3 with speaker diarization
Speaker Identification	Maps audio segments to named employees or labels unknown speakers automatically
AI Summarization	Generates structured meeting summaries, action items, decisions, and follow-ups via LLM
Auto Distribution	Sends summary emails to all meeting attendees within 5 minutes post-meeting
Admin Panel	Web-based control center for employee onboarding, policy management, and analytics
Edge-Case Resilience	Handles tab switching, multi-monitor sessions, partial participation, and reconnects
Privacy Controls	Per-employee consent, meeting exclusion lists, and data retention policies

02 System Architecture Overview

The platform is composed of four distinct layers that work together in a coordinated pipeline:

2.1 Architectural Layers

Layer	Component	Responsibility
Layer 1	Desktop Agent (.exe)	Runs on employee machines; detects meetings, captures audio, manages session state
Layer 2	Ingestion API	Receives audio/transcript chunks from agents; queues them for processing
Layer 3	AI Processing Pipeline	Transcription → Diarization → Speaker Mapping → Summarization → Action Extraction
Layer 4	Admin Panel + DB	Employee registry, meeting logs, consent management, analytics dashboard

2.2 High-Level Data Flow

The end-to-end flow from meeting detection to email delivery:

1. Desktop Agent detects an active meeting session on the employee machine (Google Meet or Teams).
2. Agent begins capturing the system audio output in real-time using WASAPI loopback capture.
3. Simultaneously, the agent polls the Microsoft Graph API (for Teams) or uses URL/process signals (for Meet) to retrieve participant metadata.
4. Audio chunks (30-second windows) are streamed to the Ingestion API over a secure WebSocket.
5. The AI Pipeline picks up each chunk, runs Whisper Large V3 Turbo for transcription, and then performs speaker diarization using pyannote.audio.
6. Post-meeting, the full transcript is assembled, speaker segments are mapped to known employees by voice-print matching, and the LLM generates the meeting summary.
7. The notification service fetches participant emails (from Graph API for Teams; from Google Calendar API for Meet) and sends the summary email.
8. All data is logged to the Admin Panel database for audit, search, and analytics.

> ℹ **NOTE** Teams meetings can also deliver a complete post-meeting transcript via Microsoft Graph API (callTranscript resource), which serves as a high-fidelity fallback or primary source, bypassing the need for audio capture entirely.

03 Desktop Agent — Architecture & Design

The Desktop Agent is a Windows application (.exe) built with Electron (cross-platform capable but primarily targeting Windows 10/11) or alternatively as a native C# WPF application. It runs as a background system tray process — invisible to the user during normal operation.

3.1 Setup & Activation Flow

- Employee downloads and installs the .exe from the Admin Panel portal link.
- On first launch, the employee enters their unique 8-character alphanumeric Activation Code (generated by admin during onboarding).
- The agent authenticates this code against the backend API, receives its configuration bundle (employee ID, org ID, feature flags, API endpoints), and stores them locally in an encrypted config store.
- The agent registers the machine and starts the background session monitor. Auto-launch on Windows startup is configured via registry entry.

> ⚠ **IMPORTANT** The activation code is single-use and bound to an employee ID. If an employee reinstalls on a new machine, admin must regenerate the code. This prevents unauthorized agent deployments.

3.2 Meeting Detection Engine

This is the most critical and technically challenging component. The agent must accurately detect when a meeting is live, which platform it is on, and avoid false positives from browser tab switching.

Detection Strategy — Microsoft Teams

- Teams runs as a native Windows process (ms-teams.exe or Teams.exe for the new Teams client).
- Primary: Monitor the process list. When ms-teams.exe is active AND the Teams window title contains keywords like "In meeting" or "Meeting in progress", treat as an active meeting.
- Superior fallback: Subscribe to Microsoft Graph API Change Notifications for onlineMeeting events tied to the employee's calendar. This gives a webhook-based, definitive start/end signal.
- Audio activity monitor: Use WASAPI to listen for non-silent audio output from the Teams process specifically, as an additional signal.

Detection Strategy — Google Meet

- Google Meet runs inside a Chrome or Edge browser tab (meet.google.com/* URL pattern) or as a Progressive Web App (PWA).
- Monitor active browser window titles. Chrome exposes the current page title in the window title bar (format: "Call Name - Google Meet - Google Chrome").
- Use the Windows Accessibility API (UI Automation) or Native Messaging extension to read active tab URLs from Chrome/Edge.
- Match against regex pattern: https://meet.google.com/[a-z]{3}-[a-z]{4}-[a-z]{3}
- A Google Chrome Native Messaging Host (small companion extension, optionally deployed via enterprise policy) provides the most reliable URL detection method.

Session State Machine

The agent uses a formal state machine to avoid false positives:

State	Trigger	Action
IDLE	Default state	Polls for meeting signals every 5 seconds
DETECTING	Process/URL signal detected	Waits 8 seconds to confirm — prevents false positives from brief tab visits
ACTIVE	Confirmed meeting (8s threshold)	Begins audio capture, starts session timer, logs session start
PAUSED	No audio for 60+ seconds	Suspends capture; does not end session. Resumes on audio activity
ENDING	Meeting signal lost	Waits 45 seconds before declaring meeting ended (handles brief disconnects)
COMPLETED	Session confirmed ended	Uploads final audio chunk, triggers post-processing pipeline

3.3 Handling the Critical Edge Cases

Edge Case 1: Tab Switching During Google Meet

Problem: Employee switches to another browser tab during a Meet call. The window title changes, potentially triggering a false "meeting ended" event.

Solution: The agent does NOT rely solely on window title visibility. It uses two additional signals:

- Audio activity: If the system audio output from Chrome is still active (detected via WASAPI), the meeting is considered ongoing regardless of which tab is focused.
- URL persistence: The Native Messaging extension tracks the tab, not the window. Even if the Meet tab is backgrounded, its URL remains active in the tab registry.
- Grace window: The ENDING state has a mandatory 45-second grace window before the session is closed. Tab switches are typically far shorter.

Edge Case 2: Dual Meetings or Multiple Platforms Simultaneously

Problem: Employee has a Teams call open while a Google Meet tab is also in-call state. The system could log two concurrent meetings.

Solution: The agent enforces a single active session policy per machine. When two meeting signals are detected simultaneously:

- Precedence rule: Teams (native app) takes precedence over browser-based Meet.
- Admin-configurable: This precedence can be flipped in the Admin Panel per team/role.
- Notification: A system tray toast notifies the employee which session is being captured.
- The second meeting is queued for manual review or ignored based on policy.

Edge Case 3: Network Drop / System Sleep Mid-Meeting

Problem: Laptop sleeps or loses network. The agent cannot upload audio, and the meeting signal may be lost on wake.

- Audio is always buffered locally first (encrypted temp file on disk, cleared after successful upload).
- On wake, the agent checks if the meeting URL/process is still active and resumes capture.

- If the meeting ended during sleep, the agent uploads the buffered audio and processes it as a completed session.
- Heartbeat timeout: If no heartbeat reaches the server for 10 minutes, the backend flags the session for manual review instead of auto-completing.

Edge Case 4: Employee Joins Same Meeting from Two Devices

Problem: Employee joins on laptop and phone. Two agents could create duplicate sessions.

- Sessions are deduplicated at the backend by matching: employee_id + meeting_url/meeting_id + overlapping timestamp window.
- If a duplicate is detected, the session with the higher audio quality score (calculated from signal-to-noise ratio) is kept as primary.

Edge Case 5: Short / Accidental Meetings (< 2 minutes)

Problem: Employee briefly joins a meeting room by accident or for a 30-second check.

- Minimum session duration threshold: 120 seconds. Sessions shorter than this are discarded and not sent to attendees.
- Configurable per organization from the Admin Panel.

04 Transcript Acquisition Strategy

The platform uses a dual-track acquisition strategy: API-first for Microsoft Teams, audio-capture-first for Google Meet.

4.1 Microsoft Teams — Graph API Track

Microsoft Graph API provides a production-grade, officially supported method to retrieve meeting transcripts as structured .vtt files with speaker names already embedded. This is the preferred method for Teams.

How It Works

- The organization's IT admin registers the MIP application in Microsoft Entra ID (Azure AD) with the following application permissions: OnlineMeetingTranscript.Read.All, OnlineMeetings.Read.All, Calendars.Read.
- A CsApplicationAccessPolicy is assigned to the registered app so it can access transcripts on behalf of users.
- MIP subscribes to Graph Change Notifications for transcript availability: POST /subscriptions with resource /users/{id}/onlineMeetings/getAllTranscripts.
- When a transcript becomes available post-meeting, Graph fires a webhook notification to MIP's backend.
- MIP fetches the transcript content via GET /users/{userId}/onlineMeetings/{meetingId}/transcripts/{transcriptId}/content.
- The .vtt transcript already contains speaker names, timestamps, and spoken language metadata — it is ingested directly into the summarization pipeline.

Microsoft Graph AI Insights (Additional Feature)

- Microsoft Teams Copilot's Meeting AI Insights API (GET /copilot/users/{userId}/onlineMeetings/{meetingId}/aiInsights) provides structured meeting notes and action items.
- MIP can optionally consume these insights to augment or validate its own LLM-generated summary.

> ⚠ **IMPORTANT** Graph API transcripts are only available for scheduled calendar-backed meetings, not ad-hoc calls created without a calendar event. For ad-hoc calls, the audio capture track is used as fallback.

4.2 Google Meet — Audio Capture Track

Google does not offer a public API for real-time meeting transcription for standard Workspace accounts. MIP therefore relies on system audio capture.

Audio Capture Mechanism

- WASAPI Loopback: On Windows, WASAPI (Windows Audio Session API) allows capturing the system audio output (what the speakers are playing) without any microphone — this captures all remote participants' voices as they come through the speakers.
- The agent opens a loopback capture session on the default audio render device, capturing at 16kHz, 16-bit mono — the optimal format for Whisper.
- Audio is segmented into 30-second chunks (matching Whisper's native context window) and uploaded to the Ingestion API.

> ⚠ **IMPORTANT** WASAPI loopback captures the full system audio mix. If the employee is playing music while in a meeting, that audio will be captured too. MIP applies a Voice Activity Detection (VAD) filter using Silero VAD before sending to Whisper to suppress non-speech segments.

4.3 Transcript Source Priority Matrix

Platform	Primary Source	Fallback Source	Speaker ID Method
MS Teams (Scheduled)	Graph API VTT Transcript	WASAPI Audio Capture	Embedded in VTT file
MS Teams (Ad-hoc)	WASAPI Audio Capture	Bot Media SDK Stream	Voice-print matching
Google Meet	WASAPI Audio Capture	Companion Extension	Voice-print matching

05 AI Processing Pipeline

The AI pipeline is the core intelligence layer. It runs entirely server-side on MIP's backend infrastructure, ensuring no heavy computation occurs on the employee's machine.

5.1 Speech-to-Text — Whisper Large V3 Turbo

After extensive evaluation, Whisper Large V3 Turbo (by OpenAI, fully open-source) is selected as the primary transcription engine.

Why Whisper Large V3 Turbo

Criterion	Assessment
Hindi + English Support	Both languages natively supported. Hindi accuracy is good; code-switching (Hinglish) handled via language detection per segment.
Accuracy	7.4% average Word Error Rate on mixed benchmarks. 10–20% improvement over Whisper Large V2.
Speed	6× faster than Whisper Large V3 full model. Processes 1 hour of audio in under 3 minutes on a single A100.
Open Source	Apache 2.0 license. No per-minute API cost. Self-hosted on MIP infrastructure.
Deployment	Runs on any GPU server. Quantized (int8) version fits on RTX 4090 with 24GB VRAM.
Multilingual	99 languages supported; automatic language detection per 30-second segment.

Configuration for Bilingual (English + Hindi) Meetings

- Language is set to auto-detect. Whisper identifies the language per chunk independently.
- For Hinglish (code-switched) content, the large-v3-turbo model handles mixed segments better than smaller models.
- Optional: Fine-tune on a corpus of internal meeting recordings in Hinglish for improved accuracy on organization-specific terminology.
- faster-whisper (CTranslate2-based) implementation is used for 4× additional inference speed with equivalent accuracy.

5.2 Speaker Diarization — pyannote.audio

Speaker diarization is the process of separating "who spoke when" in a multi-speaker audio recording.

- pyannote.audio 3.x is used — it is the current state-of-the-art open-source diarization library with pipeline: speaker segmentation → speaker clustering → speaker verification.
- The diarization output is a timeline of speaker segments: [SPEAKER_01: 00:00:05 → 00:00:23], [SPEAKER_02: 00:00:24 → 00:00:41], etc.
- These segments are aligned with Whisper's word-level timestamps to produce a speaker-tagged transcript.

Speaker Identification — Mapping Speakers to Employees

- During employee onboarding in the Admin Panel, the employee optionally records a 30-second voice sample.
- This sample is processed by pyannote's speaker verification model to generate a voice embedding (256-dimensional vector).
- Voice embeddings are stored in the employee record in the database.
- During post-processing, diarized speaker segments are matched against the voice embedding database using cosine similarity. A threshold of 0.85 cosine similarity is required for a confirmed match.
- Unmatched speakers are labeled as UNKNOWN_1, UNKNOWN_2, etc. in the output.
- For Teams meetings where Graph API VTT is available, speaker names from the VTT file override voice-print matching.

5.3 Meeting Summarization — LLM

The assembled, speaker-tagged transcript is passed to an LLM for structured summarization.

LLM Options — Recommended Stack

Option	Model	Trade-off
Primary (Cloud)	GPT-4o-mini via OpenAI API	Best quality, lowest cost (~$0.15/1M input tokens), minimal latency. Recommended default.
Open Source Self-Hosted	Llama 3.1 70B (8-bit quantized)	Zero per-meeting API cost, full data privacy. Requires 2× A100 GPUs. Best for large-volume deployments.
Lightweight Open Source	Mistral 7B Instruct / Qwen2.5 7B	Runs on single RTX 4090. Slightly lower quality on long transcripts but adequate for most use cases.

Summarization Output Schema

The LLM is given a structured prompt that produces a JSON response with these fields:

- meeting_title — inferred from conversation context
- date_time — from session metadata
- duration — in minutes
- attendees — list of names with roles
- executive_summary — 3–5 sentence overview
- key_discussion_points — bulleted list of main topics
- decisions_made — specific decisions reached, with owner
- action_items — list of tasks with owner and due date (LLM extracts from context)
- open_questions — unresolved questions that need follow-up
- next_steps — recommended follow-ups
- sentiment — overall meeting tone (constructive / tense / neutral)

✓ **BEST PRACTICE** The LLM prompt explicitly instructs the model to mark any field as "Not determinable from transcript" if insufficient context exists, rather than hallucinating content.

06 Admin Panel — Design & Features

The Admin Panel is a secure web application accessible by HR, IT, and team managers. It serves as the central control center for employee management, policy configuration, meeting logs, and analytics.

6.1 Employee Onboarding Module

Creating an Employee Account

9. Admin navigates to Employees → Add Employee.
10. Fills in: Full Name, Employee ID, Email, Department, Role/Designation, Reporting Manager, Teams Email (for Graph API mapping), Google Workspace Email (for Calendar API).
11. System generates a unique 8-character Activation Code (format: XXXX-XXXX, alphanumeric, case-insensitive).
12. Admin sends the activation email (pre-built template) to the employee with the code and download link.
13. Employee installs the agent, enters the code, and the account is linked.

Employee Record Fields

Field	Type	Purpose
employee_id	UUID	Primary key
full_name	String	Display name in summaries and emails
email	String	Primary email for summary delivery
teams_upn	String	Microsoft Teams User Principal Name for Graph API mapping
google_email	String	Google Workspace email for Calendar API participant lookup
voice_embedding	Float[]	256-dim vector from pyannote speaker verification enrollment
activation_code	String (hashed)	One-time use agent registration code
device_fingerprint	String	Registered machine identifier for security
consent_given	Boolean	Employee has acknowledged meeting recording consent
exclusion_keywords	String[]	Meeting title keywords that should NOT be captured (e.g., "1:1 HR")
role	Enum	Employee / Manager / Admin — controls what data they can view

6.2 Policy & Configuration Management

- Minimum meeting duration (default: 2 minutes) — below this, session is discarded.
- Maximum recording duration (default: 4 hours) — safety cap for runaway sessions.
- Blackout periods — time windows where no recording occurs (e.g., 6 PM–9 AM).
- Meeting title exclusions — regex patterns for meeting names that should not be recorded (e.g., "interview", "personal").
- Data retention period — how long transcripts and audio are kept (default: 90 days; configurable).

- Email delivery delay — minutes after meeting end before summary is sent (default: 5 minutes, allowing for any processing).
- Language override — force transcription in English only (for orgs that do not need Hindi output).

6.3 Meeting Logs & Search

- Every captured meeting is logged with: session ID, platform, employee, date/time, duration, participant count, transcript preview, summary, processing status.
- Full-text search across all transcripts within the retention window.
- Filter by employee, department, date range, meeting platform, duration.
- Admin can view the full transcript and generated summary for any meeting.
- Transcript correction: Admin or employee can annotate corrections to the transcript (for training data improvement).

6.4 Analytics Dashboard

- Meetings per employee per week — workload distribution view.
- Average meeting duration by department — identifies departments with meeting overhead.
- Action item completion rate — if integrated with task management tools.
- Top recurring discussion topics — keyword frequency analysis across all transcripts.
- Platform breakdown — Teams vs. Google Meet usage split.
- Failed/missed captures — sessions where the agent failed to capture or process successfully.

07 Email Distribution System

Post-meeting emails are the primary user-facing deliverable. They must be reliable, well-formatted, and reach the right people every time.

7.1 Attendee Identification

Platform	Attendee Discovery Method
Microsoft Teams	Graph API: GET /onlineMeetings/{id}/attendanceReports — returns all attendees with email, join time, and leave time.
Google Meet	Google Calendar API: GET /calendars/{calendarId}/events?q={meetingUrl} — retrieves calendar event attendees. Or Google Workspace Admin Reports API for Meet audit logs.
Unknown Attendees	External guests identified in transcript as UNKNOWN are listed in the email with a note that speaker could not be identified.

7.2 Email Content Structure

The summary email uses a clean HTML template with the following sections:

- Header: Meeting title, date, time, duration, platform icon (Teams/Meet logo).
- Attendees: Comma-separated list of participants. Present / Joined Late / Left Early flags where data is available.
- Executive Summary: 3–5 sentences.
- Key Discussion Points: Bulleted list.
- Decisions Made: Table with Decision, Owner, Date.
- Action Items: Table with Task, Owner, Due Date, Priority.
- Open Questions: Bulleted list.
- Footer: "View full transcript" link (authenticated, time-limited link to Admin Panel), "Report an error" link, unsubscribe option.

7.3 Email Sending Infrastructure

- Primary: SendGrid / AWS SES for reliable transactional email delivery with delivery tracking.
- Retry logic: Failed deliveries are retried 3× with exponential backoff.
- Bounce handling: Bounced emails are flagged in Admin Panel for address correction.
- External attendee policy: By default, external (non-employee) attendees do NOT receive the summary email. Admin can whitelist specific external domains.

08 Enhanced Features & Roadmap

Beyond the core functionality, the following features significantly increase the platform's value and are recommended for phased implementation.

8.1 Phase 1 Enhancements (Launch + 30 Days)

Feature	Description
Smart Action Tracking	Action items extracted by LLM can be pushed directly to Jira, Asana, or Microsoft To Do via integrations.
Meeting Health Score	Each meeting is scored on: time overrun, participation balance, decision density, and follow-through rate.
Real-time Transcript View	Admin Panel shows a live rolling transcript during active meetings (read-only, for managers with access permission).
Consent Reminder Flow	If a meeting participant has not given consent, the organizer receives an automated reminder before the meeting starts.
Keyword Alerts	Admin configures keywords (e.g., "escalation", "legal", "urgent"). When mentioned in a meeting, an instant Slack/email alert is sent to the designated stakeholder.

8.2 Phase 2 Enhancements (90 Days)

Feature	Description
Multi-language Support	Extend Whisper pipeline to support additional Indian languages: Tamil, Telugu, Kannada, Marathi (Whisper Large V3 supports all of these).
Meeting Series Intelligence	For recurring meetings, the LLM compares this week's summary against previous summaries to highlight what changed, what was revisited, and what was never resolved.
Personal Meeting Dashboard	Each employee gets a self-service portal showing their own meeting history, their action items, and their "meeting load" over time.
Voice Enrollment via Self-Service	Employees can record their voice enrollment sample directly from the personal portal — no admin intervention needed.
Salesforce CRM Integration	For sales calls, automatically create/update Salesforce opportunities, contacts, and activity logs from meeting summaries.

8.3 Phase 3 Enhancements (6 Months)

Feature	Description
Coaching Insights	For sales or client-facing teams: talk-time ratio, question-to-statement ratio, filler word frequency, sentiment shift during objection handling.
Meeting Library & Search	Semantic search across all past meeting transcripts using vector embeddings. "Find all meetings where we discussed the Q4 budget."
Cross-meeting Knowledge Graph	Build a graph of topics, decisions, people, and action items across all meetings to surface organizational knowledge.
Mac OS Agent	Extend the desktop agent to macOS for organizations using Apple hardware. Uses CoreAudio for audio capture.

Feature	Description
Zoom Integration	Add Zoom meeting detection and transcript acquisition via Zoom Meeting SDK or Webhook API.

09 Technology Stack

9.1 Desktop Agent

Component	Technology	Rationale
App Framework	Electron (Node.js + Chromium) or C# WPF	Electron for rapid cross-platform development; WPF for lighter Windows-native footprint
Audio Capture	NAudio (C#) / WASAPI Loopback	Native Windows audio API; captures system audio without microphone access
Meeting Detection	Windows UI Automation API + Process Monitor	Reliable native detection without browser extension dependency
Browser Extension	Chrome/Edge Native Messaging Host	Provides URL-level tab tracking for Google Meet
Secure Config	Windows DPAPI + encrypted SQLite	Local credential storage; tamper-resistant on the machine
Update Mechanism	Electron-updater / Squirrel.Windows	Silent background auto-updates; no employee action required

9.2 Backend & AI Infrastructure

Component	Technology	Rationale
API Server	FastAPI (Python)	Native Python ecosystem for AI/ML libraries; async support for WebSocket audio streaming
Task Queue	Celery + Redis	Distributes audio chunk processing jobs across worker nodes; handles backpressure
STT Engine	faster-whisper (CTranslate2) — Whisper Large V3 Turbo	Fastest open-source multilingual ASR; 4–6× faster than original Whisper implementation
Diarization	pyannote.audio 3.x	State-of-the-art open-source speaker diarization and verification
VAD	Silero VAD	Lightweight, fast voice activity detection to filter non-speech segments
LLM Summarization	GPT-4o-mini (primary) / Llama 3.1 70B (self-hosted)	Cost-efficient cloud model with self-hosted fallback for data-sensitive orgs
Database	PostgreSQL + pgvector	Relational data for all records; pgvector for voice embedding similarity search
Cache	Redis	Session state, rate limiting, job queuing
Object Storage	AWS S3 / Azure Blob	Encrypted audio file storage during processing; auto-deleted after retention period
Email	SendGrid / AWS SES	Transactional email with delivery tracking and bounce management

9.3 Admin Panel

Component	Technology	Rationale
Frontend	React + TypeScript + Tailwind CSS	Modern, maintainable UI stack; rapid component development
Charts & Analytics	Recharts / Chart.js	Lightweight charting for meeting analytics dashboard
Auth	Clerk / Auth0 with SSO support	Enterprise SSO via Azure AD / Google Workspace; MFA enforced for admin roles
Hosting	Vercel (frontend) + AWS ECS (backend)	Scalable, managed infrastructure with auto-scaling
Infrastructure	Docker + Kubernetes (EKS)	Containerized microservices; independent scaling of AI worker nodes

10 Security, Privacy & Compliance

Meeting recordings contain sensitive organizational information. Security and privacy controls must be architected from day one — not bolted on.

10.1 Data Security

- All audio in transit: TLS 1.3 encrypted WebSocket connections from agent to API.
- All audio at rest: AES-256 encrypted in S3/Blob storage. Encryption keys managed by AWS KMS / Azure Key Vault.
- Audio is deleted from object storage after the AI pipeline completes processing. Transcripts, not raw audio, are the long-term record.
- Admin Panel is protected by MFA and role-based access control (RBAC). Managers can only see meetings involving their direct reports.
- Agent-to-API authentication: JWT tokens signed with RS256. Tokens are rotated every 24 hours.
- Activation codes are hashed (Argon2) in the database. The plaintext code is never stored.

10.2 Employee Privacy & Consent

- Consent-first: No employee recording begins until the employee has accepted the consent acknowledgment in the desktop agent.
- Opt-out: Employees can disable capture for any meeting via the system tray icon. This is logged but not penalized.
- Exclusion keywords: Employees can add their own personal meeting exclusion keywords (e.g., "therapy", "personal") that are processed only on their device and never sent to the server.
- Data subject rights: Employees can request deletion of all their recorded meetings from the Admin Portal (GDPR/DPDPA compliance).
- Transparency: A visible recording indicator is shown in the system tray whenever a session is actively captured.

10.3 Compliance Considerations

> ⚡ **RISK** This platform potentially falls under GDPR (EU), DPDPA (India), and various state-level privacy laws. Legal review is mandatory before deployment. Key actions: publish a clear employee privacy policy, establish data processing agreements with all third-party vendors (OpenAI, SendGrid, AWS), implement data residency controls for regulated industries.

- Indian organizations: Compliance with India's Digital Personal Data Protection Act (DPDPA) 2023 — consent requirements, data retention limits, and data localization may apply.
- Regulated sectors (banking, healthcare): Consider fully self-hosted deployment with no data leaving the organization's cloud environment.
- Legal hold: Admin Panel should support placing a meeting on legal hold (exempt from automatic deletion) when triggered by the legal team.

11 Deployment & Infrastructure Plan

11.1 Backend Infrastructure — GPU Requirements

Scale	GPU Requirement	Notes
< 50 concurrent meetings	Single NVIDIA RTX 4090 (24GB)	Runs Whisper Large V3 Turbo + pyannote on one GPU. Sufficient for SMB.
50–200 concurrent meetings	2–4× A10G or RTX 4090	Horizontal scaling via Kubernetes GPU node pool. Celery worker per GPU.
200+ concurrent meetings	A100 cluster	Enterprise deployment. Consider AWS p4d instances or Azure NC A100 v4.

11.2 Phased Rollout Plan

Phase	Timeline	Deliverables
Phase 0	Weeks 1–4	Core infrastructure setup: Backend API, DB schema, Admin Panel skeleton, Whisper + pyannote pipeline proof of concept.
Phase 1	Weeks 5–8	Desktop Agent (Teams detection + WASAPI capture), Graph API Teams transcript integration, Admin Panel employee onboarding, basic summary emails.
Phase 2	Weeks 9–12	Google Meet detection (Native Messaging), voice enrollment, speaker diarization pipeline live, analytics dashboard, policy management.
Phase 3	Weeks 13–16	Action item integrations (Jira/Asana), keyword alerts, meeting health scores, edge case hardening, performance testing at scale.
Phase 4	Weeks 17–20	Beta rollout to pilot group (10–20 employees), feedback loop, accuracy fine-tuning, security audit, production launch.

12 Risk Register

Risk	Likelihood	Impact	Mitigation
Microsoft Graph API changes behavior or requires new permissions	Medium	High	Maintain WASAPI audio capture as permanent fallback. Monitor Graph changelog.
Hindi/Hinglish accuracy below threshold for client use	Medium	Medium	Fine-tune Whisper on internal recordings. Offer manual correction UI.
Employee resistance to recording consent	High	High	Clear communication campaign. Strong opt-out controls. Manager-only visibility policies.
WASAPI loopback blocked by enterprise audio policy	Low	High	Test in target org environment early. Alternative: virtual audio cable injection.
Speaker diarization quality degrades in large meetings (10+ speakers)	Medium	Medium	pyannote has known limitations beyond 8 speakers. Use Graph API speaker names for Teams.
Data breach of meeting transcripts	Low	Critical	AES-256 at rest, TLS in transit, RBAC, audit logging, penetration testing pre-launch.
LLM hallucination in action items	Medium	Medium	Prompt engineering with explicit anti-hallucination instructions. User verification UI.
Google changing Chrome extension policies	Low	Medium	Window title + audio fallback detection does not require extension. Extension is additive only.

13 Open Decisions Required

The following decisions need stakeholder input before development can begin on those specific components:

14. LLM Hosting: Cloud (OpenAI API) vs. fully self-hosted (Llama 3.1 70B)? Self-hosted has zero ongoing API cost but requires significant GPU capital expenditure.
15. Desktop Agent Technology: Electron (faster dev, cross-platform) vs. C# WPF (lighter, more Windows-native, easier WASAPI integration)?
16. Employee Consent Model: Opt-in (employee actively enables) vs. Opt-out (enabled by default, employee can disable)? Legal team must advise.
17. External Guest Email Policy: Should external meeting guests receive the summary email by default?
18. Data Residency: Must all data remain within India, or is US-based AWS/Azure acceptable?
19. Voice Enrollment: Mandatory or optional? Mandatory improves speaker ID accuracy but may face employee resistance.
20. Microsoft Teams License Requirement: Graph API transcript features require Microsoft 365 E3/E5 or Teams Premium. Confirm all target organizations have this.

14 Glossary

Term	Definition
Diarization	Process of partitioning an audio stream into segments according to the speaker identity ("who spoke when").
VAD	Voice Activity Detection — determines when speech is present in an audio signal, filtering out silence and noise.
WASAPI	Windows Audio Session API — Microsoft's low-level audio API; loopback mode captures system audio output.
Graph API	Microsoft's REST API for accessing Microsoft 365 services, including Teams meetings and transcripts.
Whisper	OpenAI's open-source multilingual speech recognition model, trained on 680,000 hours of audio.
pyannote	Open-source Python library for speaker diarization and verification, built on PyTorch.
WER	Word Error Rate — standard metric for speech recognition accuracy. Lower is better.
VTT	WebVTT format — standard subtitle/caption file format used by Microsoft's Graph API for transcript delivery.
RSC	Resource-Specific Consent — Microsoft's model for granting app permissions scoped to a specific Teams meeting or chat.
DPAPI	Windows Data Protection API — encrypts local data using the logged-in user's credentials as the key.
MIP	Meeting Intelligence Platform — this system.

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